Fundamentals of System Security
(COMSM0122)

Group Coursework (40 Marks)
Engineering and Securing a National Emergency Response Platform (NERP)

Submission Deadline: Friday 6th November 2026 @16.00

Introduction

Modern emergency response systems are among the most critical digital systems operated by a nation. Their failure can delay life-saving interventions, disrupt public safety operations, compromise sensitive intelligence, and undermine public trust in government institutions. This coursework places your team in the role of a security engineering consultancy commissioned by the UK Government to design, implement, secure, and evaluate a next-generation emergency-response platform known as the National Emergency Response Platform (NERP). The platform must coordinate police, ambulance, fire and rescue services, intelligence analysts, regional command centres, and national crisis coordinators. It must remain operational during infrastructure failures, cyber-attacks, regional outages, communication disruptions, and insider threats.

Your objective as a team is to engineer a platform that remains trustworthy when assumptions fail. Throughout the coursework you will be required to analyse the trade-offs between confidentiality, integrity, availability, operational practicality, scalability, and trust. You will investigate how distributed systems preserve agreement, how information should flow through an organisation, and how access to sensitive resources should be controlled to preserve trust under adverse conditions.

Learning Outcomes

On successful completion of this coursework, students should be able to demonstrate the ability to design secure distributed systems, evaluate consensus protocols and their assumptions, engineer confidentiality and integrity controls, implement access-control mechanisms, perform adversarial security analysis, evaluate security trade-offs, and construct evidence-based security assurance arguments.

More importantly, students should be able to reason about the relationship between system correctness and security, recognising that many failures emerge not from incorrect algorithms but from violations of the assumptions upon which those algorithms depend.

Scenario

The UK Government has initiated a programme to replace a collection of fragmented regional emergency-response systems with a single nationwide platform. The National Emergency Response Platform (NERP) will provide emergency call handling, incident coordination, resource allocation, intelligence distribution, operational auditing, and analytical reporting. Multiple agencies must coordinate using the system while maintaining both security and operational effectiveness. Information processed by the platform ranges from publicly releaseable emergency alerts to highly classified

national-security intelligence. The platform operates across multiple regions and datacentres. During major incidents, communication links may fail, infrastructure may become unavailable, and parts of the system may be actively targeted by attackers.
Your consultancy team has been asked to design and implement a prototype demonstrating how NERP could provide resilient, secure, and trustworthy operation under these conditions.

System Requirements

Your implementation must contain a collection of interacting services that collectively simulate the core capabilities of the platform. At minimum, the platform should include services responsible for authentication, intelligence management, emergency dispatch, incident management, auditing, analytics, information governance, and distributed consensus.

A typical implementation might contain modules such as:

```
nerp/
├── auth/
├── incidents/
├── dispatch/
├── intelligence/
├── analytics/
├── audit/
├── datastore/
├── consensus/
│   ├── bft/
│   ├── paxos/
│   └── raft/
├── policy-engine/
└── experiments/
```

You may extend or modify the architecture where appropriate. Any modifications should be justified within the report.

Part A – Architecture and Threat Modelling (4 Marks)

This section establishes the overall security foundation for the platform. The intention is to demonstrate an understanding of how architecture influences security, resilience and operational effectiveness.

Task A1 – Distributed Architecture Design (2 Marks)
(Maximum: 250 words – Diagrams excluded from word count)

Design a complete architecture for NERP. Your architecture should clearly identify the major services within the system, the relationships between them, and the mechanisms used to support availability, confidentiality and integrity.
The architecture should explicitly identify trust boundaries, security domains, replication mechanisms, datastores, authentication services and audit services. It should also explain how the system is distributed across multiple regions and how failures are tolerated. The objective is not to produce a diagram alone. You should explain the security rationale behind your design decisions and describe how those

decisions affect operational behaviour. At least three trust boundaries and two security domains must be identified.

Task A2 – Threat Model (1 Mark)
(Maximum: 250 words – Threat tables excluded from word count)
Develop a threat model for the proposed architecture. At a minimum, your threat model should consider denial-of-service attacks, credential theft, insider abuse, malware, supply-chain compromise, datacentre failure, network partitions and data corruption. For each threat, identify the assets at risk, describe the attack path, discuss likely consequences, and explain how the proposed architecture mitigates the threat.

The strongest submissions will demonstrate clear links between architectural decisions and threat mitigation.

Task A3 – Architecture Assurance Summary (1 Mark)
(Maximum: 150 words)
Explain why the proposed architecture is capable of supporting NERP's availability, confidentiality and integrity requirements.

Part B – Consensus Engineering and Security Assessment (10 Marks)
Distributed consensus lies at the heart of NERP. In this unit, you explored Byzantine Fault Tolerance, Paxos and Raft. This section requires you to move beyond protocol mechanics and examine the assumptions upon which consensus depends.

The objective is not merely to demonstrate that consensus works under normal conditions. Rather, the objective is to demonstrate how and why consensus fails when trust assumptions are violated.
Workshop implementations are provided as educational reference material. Students may reuse, adapt, extend or replace these implementations. Assessment will focus on security analysis, experimentation, hardening, justification and assurance arguments rather than implementation size or complexity.

Task B1 – Consensus Selection (1 Mark)
(Maximum: 350 words - Comparison tables excluded from word count)
NERP contains several distributed subsystems, including the Regional Incident Database, Emergency Dispatch Service, Resource Allocation Service, National Audit Log and Intelligence Distribution Service. For each subsystem, select an appropriate consensus protocol from Byzantine Fault Tolerance, Paxos or Raft.

Your analysis should justify why a protocol is appropriate for a particular subsystem and explain why alternative protocols were rejected. Discussion should consider operational complexity, trust assumptions, attack surfaces, availability requirements and performance considerations.

Marks will be awarded for quality of reasoning rather than for selecting a particular protocol. Note: You are only required to test and analyse one protocol per subsystem.

Task B2 – Consensus Mental Models (2 Marks)
(Maximum 600 words – Approximately 200 words per protocol)

A central objective of the unit has been the development of mental models for reasoning about distributed agreement.

(1) For Byzantine Fault Tolerance, explain the statement: *Trust the protocol, not the participants.*
Your discussion should consider protocol evidence, validator identities, cryptographic signatures and quorum formation.

(2) For Paxos, explain the statement: *A majority is the system's memory.*
Your discussion should consider quorum overlap, value preservation, persistent state and protocol safety.

(3) For Raft, explain the statement: *Leadership before consensus.*
Your discussion should consider authority, elections, leadership, log replication and consistency.

The intention is not to describe protocol phases, but to explain the security reasoning that underpins protocol correctness.

Task B3 – Security-Critical Assumptions (2 Marks)
(Maximum 600 words – Approximately 200 words per protocol)
Consensus protocols depend upon assumptions regarding trust, storage, communication and system behaviour. Identify and explain the assumptions required by each protocol selected in Task B1.

Your analysis should consider issues such as validator authenticity, key management, membership management, quorum overlap, durable storage, proposal ordering, election timing and communication reliability. For each assumption, explain why the protocol depends upon it and discuss the consequences of violating that assumption.

The strongest submissions will identify assumptions beyond those explicitly discussed during lectures and workshops.

Task B4 – Failure and Attack Analysis (2 Marks)
(Maximum 600 words – The report should summarise the experiments only, and detailed logs and evidence belong in the Security Assurance Portfolio.)
A security engineer must understand a protocol's behaviour when assumptions fail. You are required to implement and evaluate deliberate failures and attacks against consensus systems.

(1) The Byzantine Fault Tolerance component must demonstrate an equivocation attack, a Sybil attack and a validator key-compromise scenario.
(2) The Paxos component must demonstrate state loss, majority loss and quorum misconfiguration.
(3) The Raft component must demonstrate network partitioning, election storms and leadership churn.

For each experiment, explain the assumption being violated, the modifications introduced, the behaviour observed, the affected security properties and any mitigations that may be applied.

Evidence generated during this task must be included within the Security Assurance Portfolio.

Task B5 – Original Hardening Mechanism (1 Mark)
(Maximum 250 words)
Design and implement one security enhancement that was not provided as part of the workshop material. The enhancement should address a realistic threat or operational weakness and should strengthen one or more assumptions identified in Task B3. The hardening mechanism may be applied to any consensus protocol examined in this coursework.

Your submission should explain the problem being addressed, describe the implementation, evaluate its effectiveness and discuss any remaining limitations. Originality, justification and quality of evaluation will be considered during marking.

Task B6 – Consensus Assurance Argument (2 Marks)
(Maximum 250 words)
Using the evidence from Tasks B1–B5, evaluate whether the consensus mechanisms selected for NERP can be trusted to support a national emergency-response platform. Reflect on the failures and attacks you implemented, the assumptions they violated, and the effectiveness of any mitigations or hardening mechanisms introduced.

Your discussion should identify the most important assumptions that remain critical to safe operation, any residual risks that cannot be eliminated, and whether you would recommend continued use of the selected protocols within NERP. The objective is to demonstrate that trustworthy consensus depends not only on protocol correctness, but also on protecting the assumptions upon which that correctness relies.

Part C – Information Classification and Data Governance (5 Marks)
The NERP platform processes information originating from multiple agencies and operational environments. To support this coursework, a set of source datasets has been provided. These datasets represent a small snapshot of information managed by the platform and contain a mixture of operational, personal, intelligence, and audit-related data.

The datasets provided are:

```
data/

├── incidents.json
├── responders.json
├── intelligence.json
├── dispatch.json
└── audit.json
```

The records intentionally contain data of varying sensitivity and operational value. Some information may be suitable for broad internal use, while other information may require strict confidentiality and access control. Your task is to design governance mechanisms that enable the organisation to use the data effectively while reducing unnecessary security and privacy risks.

Task C1 – Classification Model (1 Mark)
(Maximum 250 words)
Design an information-classification scheme suitable for NERP and apply it to the provided datasets. Your classification model should support operational decision-making while protecting sensitive information from inappropriate disclosure. The model must explain how different information types found within the datasets should be classified and why.
The classification decisions should be justified with reference to confidentiality requirements, operational needs, and potential harm resulting from unauthorised disclosure.

Task C2 – Classification Engine (1 Mark)
(Maximum 250 words)
Implement an automated classification mechanism that assigns security classifications to records contained within the supplied datasets.

Your implementation should analyse record content and metadata when determining classifications. Examples of factors that may influence classification include the presence of personally identifiable information, medical information, operational details, intelligence indicators, or national-security content. Your report should describe the classification logic, explain how the system handles ambiguous records, and discuss any limitations of the approach.

Task C3 – Information Flow Engineering (2 Marks)
(Maximum 300 words)
Different stakeholders require different views of the same information. Using the provided datasets, implement mechanisms that generate:

- An Operational View containing information required by emergency responders.
- An Analytics View containing information suitable for operational analysis.
- A Research View containing appropriately sanitised or anonymised information.

Your implementation should support classification, sanitisation, anonymisation, archiving and secure destruction of information. Demonstrate how data flows between views while preserving appropriate confidentiality and integrity guarantees. The report should explain which fields were retained, transformed or removed and justify those decisions.

Task C4 – Retention and Governance (1 Mark)
(Maximum 250 words)
Design and implement retention policies for the information contained within the supplied datasets. The retention strategy should balance legal obligations, operational utility, security requirements and storage constraints.

Your report should explain how retention decisions vary across information classifications and demonstrate how those policies are enforced by your implementation.

Part D – Multi-Layer Security Policy Engine (6 Marks)

Emergency-response systems must balance operational effectiveness with strong security controls. During a major incident, personnel require rapid access to information, but unrestricted access can expose sensitive intelligence, compromise investigations, or allow inaccurate information to influence operational decisions.

The purpose of this section is to move beyond implementing individual security models and instead engineer a policy framework that supports secure decision-making within NERP. You will use the datasets supplied as part of the coursework to design, implement and evaluate a policy architecture that combines operational access requirements with confidentiality and integrity protections.

The policies designed and implemented in this section must be applied to the datasets provided in Part C.

Task D1 – Access Control Policy Design (1 Mark)
(Maximum 250 words – Policy matrices excluded from word count)

Before implementing any access-control mechanism, analyse the operational roles and responsibilities present within NERP.

Using the supplied datasets, identify which users require access to which resources and justify those requirements. Based on your analysis, design and implement an Access Control Matrix that defines the permissions available to roles such as dispatchers, police officers, paramedics, fire officers, intelligence analysts, regional commanders, national commanders and auditors.

Your policy should be derived from operational requirements within the NERP scenario rather than from arbitrary assumptions. The implementation should support meaningful access decisions across the incident, dispatch, intelligence and audit datasets.

Task D2 – Confidentiality Policy Engineering (1 Marks)
(Maximum 250 words)

NERP processes information ranging from public emergency notifications to highly classified intelligence reports. Not all information should be available to all personnel. Using the supplied datasets, identify which information assets require confidentiality protection and justify the classification levels assigned to them.

Implement confidentiality controls using the Bell-LaPadula model and demonstrate how those controls prevent inappropriate information disclosure. Your implementation should illustrate the practical application of the principles commonly described as *No Read Up* and *No Write Down*.

Your discussion should focus on how confidentiality supports emergency operations and what risks emerge if classification boundaries are violated.

Task D3 – Integrity Policy Engineering (1 Mark)
(Maximum 250 words)

Operational decisions within NERP rely upon trusted information. Incorrect casualty figures, manipulated intelligence reports, altered dispatch orders or compromised audit logs could have significant consequences during an emergency.

Using the supplied datasets, identify the information assets whose integrity is most critical to NERP operations and justify your choices.
Implement integrity controls using the Biba model and demonstrate how those controls preserve trustworthiness through the practical application of *No Read Down* and *No Write Up*.

Your discussion should explain how integrity violations could affect emergency response activities and why different information assets may require different levels of protection.

Task D4 – Policy Conflict and Security Evaluation (2 Marks)
(Maximum 300 words)
Security models often have competing objectives. A request that appears reasonable from an operational perspective may conflict with confidentiality requirements, while a confidentiality policy may permit an action that introduces integrity risks.
Integrate the Access Control Matrix, Bell-LaPadula and Biba models into a single policy engine and use the supplied datasets to evaluate its behaviour.

Demonstrate at least three scenarios in which:

1. The Access Control Matrix permits an operation but Bell-LaPadula prevents it.
2. Bell-LaPadula permits an operation but Biba prevents it.
3. All security controls permit the operation.

For each scenario, explain why the final decision is appropriate and identify the security property being protected.
The objective of this task is not merely to grant or deny access, but to demonstrate how confidentiality, integrity and operational effectiveness interact within a secure emergency-response platform.

Task D5 – Access Control Assurance Argument (1 Marks)
(Maximum 250 words)
Using the evidence generated in Tasks D1–D4, evaluate whether the policy architecture implemented for NERP provides an appropriate balance between operational effectiveness and security.
Identify the assumptions upon which your policy decisions depend, discuss any residual risks that remain, and explain how confidentiality, integrity and operational requirements were balanced throughout the design.

Conclude by explaining whether you believe the resulting policy framework is suitable for a national emergency-response platform and justify your position using evidence from your implementation and experiments.

Part E – Security Governance and Assurance (8 Marks)
Security engineering frequently involves competing requirements with no perfect solution. While previous sections focused on architecture, consensus, information governance and access control, this section focuses on decision-making. You should assume that NERP has been deployed and is operating successfully. New operational demands, regulatory requirements and security incidents now place pressure on the assumptions and controls that underpin the platform.

The purpose of this section is to evaluate your ability to reason about difficult trade-offs, justify security decisions, and communicate residual risk. The strongest submissions will recognise that secure systems rarely maximise confidentiality, integrity and availability simultaneously. Instead, security architects must make informed decisions about which properties should be prioritised under specific circumstances.

Task E1 – Strategic Security Impact Assessment (3 Marks)
(Maximum 500 words)

The following changes occur after deployment:

- A research organisation requests unrestricted access to intelligence data.
- Emergency responders request direct access to classified national-security information.
- A regional datacentre becomes permanently unavailable.
- Validator credentials protecting the national audit system are compromised.
- New government regulations require all emergency-response records to be retained for twenty years.

For each scenario, assess the impact on NERP, including the supplied datasets, from a security, operational and governance perspective.

Your analysis should identify which security properties are affected, which parts of the architecture are impacted, and whether any consensus assumptions, classification controls, confidentiality policies, integrity controls or retention policies are placed under pressure.

The objective is not to propose solutions at this stage, but to determine which aspects of the current design are affected and why.

Task E2 – Security Governance Decisions (3 Marks)
(Maximum 500 words)

As the security architects responsible for NERP, you must determine how the system should respond to the changes identified in Task E1.

For each scenario, recommend a course of action and justify your decision. Your discussion should explain which requirements are being prioritised, which requirements are being sacrificed, and how the proposed response affects confidentiality, integrity, availability and operational effectiveness.

There are intentionally no perfect solutions. Marks will be awarded for the quality of reasoning, the ability to justify security decisions, and the recognition of trade-offs and residual risks.

Task E3 – NERP Security Assurance Statement (2 Marks)
(Maximum 300 words)

Drawing upon the evidence generated throughout Parts A–E and the Security Assurance Portfolio, provide a final security assurance statement for NERP.

Your statement should identify the most significant residual risks remaining within the platform, the assumptions that must remain true for safe operation, and the controls that are most important for maintaining trust in the system.

Conclude by stating whether you would recommend NERP for deployment as a national emergency-response platform and justify your position using evidence from

your architecture, consensus analysis, information-governance design, access-control framework and security experiments.

The purpose of this task is not to argue that the system is perfectly secure, but to provide a reasoned assessment of whether it is sufficiently trustworthy to operate as critical national infrastructure.

Security Assurance Portfolio (7 Marks)
Each group will also submit a Security Assurance Portfolio. The portfolio should resemble the evidence package that might be provided to a government security review board. Its purpose is to demonstrate that the system has been designed, implemented, evaluated, attacked, defended and analysed systematically.

For every attack, failure scenario, mitigation and hardening mechanism, the portfolio should provide a description of the scenario, the assumption involved, the observed behaviour, the impact on security properties, the mitigation applied and any residual risks that remain. Evidence may include architecture artefacts, system logs, screenshots, experimental outputs, attack analyses, mitigation results and security arguments. The portfolio should ultimately answer a single question: **Why should NERP be trusted to operate during a national emergency?**

The Security Assurance Portfolio is not subject to a word limit. Evidence presented in the Security Assurance Portfolio may be referenced from the report and does not need to be repeated within the report itself. Evidence should be concise but sufficiently detailed to support the claims made in the report. Portfolio contents should include:

- Architecture diagrams
- Threat tables/diagrams
- Consensus logs
- Attack evidence
- Mitigation evidence
- Screenshots
- Experimental outputs
- Access-control matrices
- Classification schemas
- Security case documentation

These artefacts should be referenced in the report and are excluded from the report word count. The assurance portfolio will be marked based on the following criteria:

Criterion	**Marks**
Architecture and Threat Evidence	1
Consensus Experiments and Logs	2
Hardening and Mitigation Evidence	1
Information Governance Evidence	1
Policy Engine Evidence	1
Overall Security Case Quality	1
Total	**7**

Deliverables
Submission should be made electronically via Blackboard as a single compressed folder named `coursework-[groupname].zip`. The submission folder should contain the following:

1. Technical Report
A single PDF document containing the group's responses to:
Part A – Architecture and Threat Modelling
Part B – Consensus Engineering and Security Assessment
Part C – Information Classification and Data Governance
Part D – Multi-Layer Security Policy Engineering
Part E – Security Governance and Assurance

The report should comply with all word limits specified within the coursework specification.

2. NERP Prototype Implementation
A compressed archive containing the complete source code, datasets, configuration files and instructions required to execute the prototype implementation.
The submission should contain all artefacts necessary to demonstrate:
- Architecture implementation
- Consensus systems
- Information governance components
- Policy engine implementation
- Security experiments
- Hardening mechanisms

3. Security Assurance Portfolio
A PDF document containing the Security Assurance Portfolio.
The portfolio should include evidence supporting the claims made throughout the report, such as:
- Architecture diagrams
- Threat models
- Consensus logs
- Attack evidence
- Mitigation evidence
- Screenshots
- Experimental outputs
- Access-control matrices
- Classification schemas
- Security case documentation

4. Individual Reflective Log
Each student must submit an individual reflective log separately via Blackboard.
The reflective log should:
- Summarise the student's contribution to the group.

- Reflect on technical, organisational and security-engineering lessons learned during the project.
- Explain how the student's understanding of system security evolved throughout the coursework.

Maximum length: 500 words.

Assessment

Submission is due by Friday 6th November 2026 @16:00.

Late submissions will be handled in accordance with the University's Code of Assessment. Information regarding late submission penalties can be found at:
University of Bristol Mark Signals and Penalties

All submitted material, including the report, source code, Security Assurance Portfolio and reflective logs, may be checked using the University's plagiarism and academic integrity monitoring systems.

For group submissions, a single member of the group may submit the shared coursework deliverables on behalf of the entire group. However, all students are individually responsible for submitting their own reflective log.

Students are responsible for verifying that all submitted files can be opened and assessed. Corrupt, incomplete or inaccessible submissions may be subject to marking penalties.

Groups: Infrastructure Security Pathway

Name	Group #
Baptiste Romain Lu Hao Ziyu Su Lassana Donzo	1
Xiyue Zhang Danish Maniyar Swapnil Prajapati Xinyan Zhang	2
Yutong Li Yifan Bei Zhaotianyi Liu Shreya Kattela	3
Vineeth Surendra Siyu Wang Zeyu Xie Pengfei Zheng	4
Yujin Liu Fangfei Cao Jia Li Theresa Abi Rached	5
Valthi Siva Ram Sankaralingam Kaleeswaran Tural Najafov Chenyihan Wang Manav Kumar	6
Mohammad Alshatti Melantho Lambrinou Neha Arunkumar Paras Tilak	7

Groups: Software Security Pathway

Name	Group #
Anas Alarbeed Penny Hsieh Nasri Ibrahim Vaibhav Tukaram Narkhede	8
Fei Yan Edward Cai Hongyi Long Chenzhi Zhao	9
Yani Luo Boguang Chen Yuan Zhuang Steven Chen	10
Lianyue Wei Steve Joseph Xinze Chen Shijie Yu	11
Yunyang Mao Yuejia Geng Yejing Feng Yudeng Zhao	12
Zhenyu Hu Ying Li Zhongao Wang Jintian Li	13
Jiashuo Li Peicheng Liu Rui Xu Xuefei Xie	14
Jinyu Xu Feihuan Zhao Mengyun Chen Sining Li	15
Jafar Ayaz Naqvi Darshan Darshan Xinyi Huang Yan Wang	16
Amaradeepan Thangamurugan Amine Ziani Wei Liu Arsyad Bin Anuar	17

Xinzhi Su Abigael Muthama Nitesh Manojkumar Tripathi Shahrad Zeinali	18
Yanhan Li Ziqin Zhao Mayurkumar Sanjaybhai Pandya Yang Jin	19